INCH-POUND
MIL-PRF-3098/90C
7 August 2009
SUPERSEDING
MIL-PRF-3098/90B
17 May 2004

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR114/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.9 MHz to 3.85 MHz; fundamental; noncontrolled; series resonance.

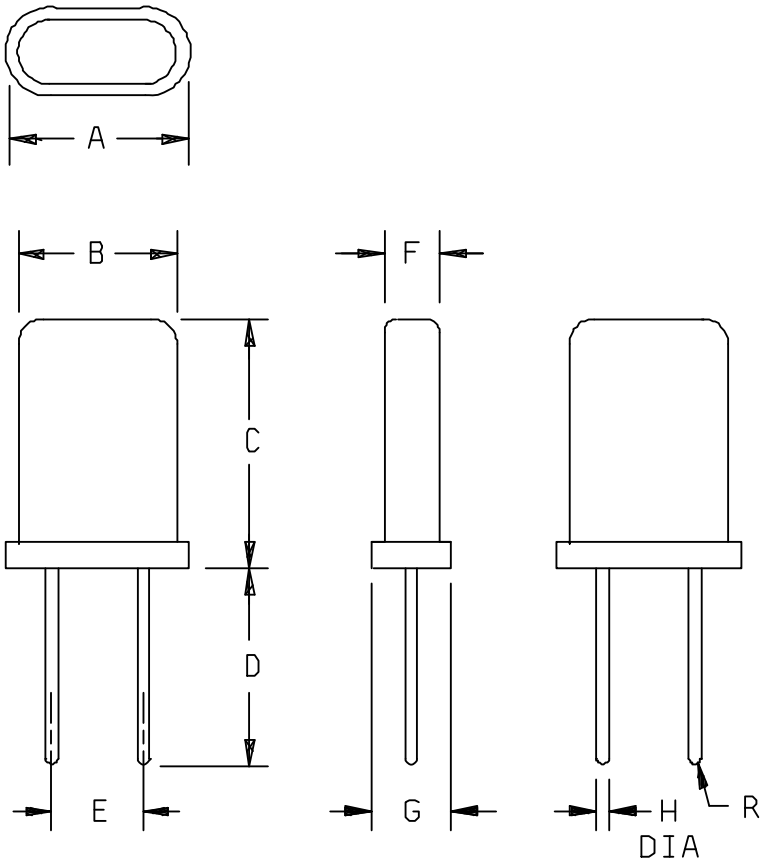

FIGURE 1. Crystal unit - CR114/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.435	---	11.05
B	---	.402	---	10.21
C	---	.530	---	13.46
D	.120	.130	3.05	3.30
E	.184	.200	4.67	5.08
F	---	.150	---	3.81
G	---	.183	---	4.65
H	.038	.042	0.96	1.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR114/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.9 MHz to 3.85 MHz, inclusive.

Frequency tolerance:

Operating temperature range: ±0.005 percent.

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Capacitance, load: 32.0 pF ±.05 pF.

Resonance: Series.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Resistance: 100 ohms.

Crystal current: 3.2 mA.

Capacitance, shunt: 7 pF, maximum.

Equivalent resistance: 300 ohms, maximum.

Shock: Method 202, MIL-STD-202.

Frequency change permitted: ±0.0005 percent.

Equivalent-resistance change permitted: ±15 percent.

Vibration: Method 204, MIL-STD-202, test condition A; .10 inch (2.54 mm) double amplitude or 5 gravity units, whichever is less.

Frequency change permitted: ±0.0005 percent.

Equivalent-resistance change permitted: ±15 percent.

Unwanted modes: Minimum unwanted mode effective resistance shall be at least twice that of the main mode, or greater, over the operating temperature range -55°C to +105°C.

Aging:

Frequency change permitted: ±0.0005 percent.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 5955-2009-033)

Review activities:
Air Force - 19, 84

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